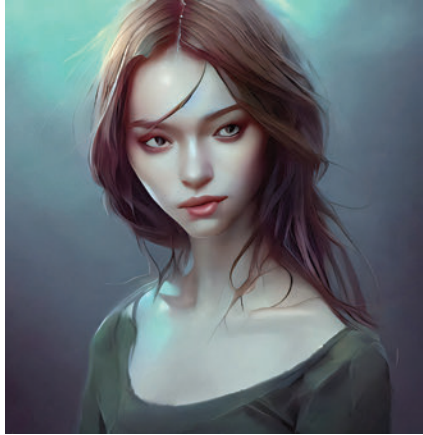


Limitations of Stable Diffusion

Despite its advanced capabilities, Stable Diffusion has limitations that can make it inferior to other image-generation frameworks in certain tasks. These limitations stem from various factors, including the limitations of the training dataset, biases introduced by the researchers, and built-in blockers to prevent misuse.

Firstly, the training data used to train the model has a significant impact on its abilities. Even with large datasets like LAION 2B(en), there is a possibility of confounding the model by referencing unseen image types with the input prompt. This means that features not included in the original training data will be impossible to recreate as the model has no understanding of them. For instance, human figures and faces are not refined features in the generated outputs as the model was largely not trained to focus on these.



Secondly, the researchers who developed Stable Diffusion recognized the inherent social bias in such a task. They trained the model primarily on subsets of the LAION-2B(en) dataset, which is almost entirely in the English language. This limits the ability to use non-English language input prompts and enables white and western cultural influences to dominate the outputted images.

Lastly, Stable Diffusion has built-in blockers that prevent the generation of abusive content. These blockers are designed to prevent the creation of harmful representations of people or their environments, cultures, religions, and other abusive content. Sharing of copyrighted or licensed material in violation of its terms of use is also prohibited.

How do text to image models work?

You've probably witnessed the impressive results of stable diffusion, let's dive into how this model operates. Stable Diffusion is built upon a diffusion model known as Latent Diffusion, which was introduced in the paper "High-Resolution Image Synthesis with Latent Diffusion Models."